

Logistic regression

Name : Siri Havaladar

Batch : August B

*Dataset : Pima Indians Diabetes Database
from Kaggle; file: diabetes.csv.*

Date : 1-9-2026

Project Objective

I developed this project to predict whether a person is likely to have diabetes based on different health-related measurements.

I used a diabetes dataset containing 768 patient records and 9 columns. The input features include Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

The Outcome column is my target variable, where 0 represents one class and 1 represents the diabetic class used by my project.

My overall machine-learning process was:

Raw patient data → Data cleaning →

Train/test preparation → Imputation →

Scaling → Logistic Regression → Prediction →

Evaluation → Coefficient interpretation

1. Replace Invalid Zeros

First, I investigated the dataset to identify values that could be invalid or could represent missing information.

I found that some columns contained zero values. For example, the original dataset had:

- 227 zero values in SkinThickness
- 374 zero values in Insulin
- 111 zero values in Pregnancies
-

The value-count analysis revealed these unusual zero values.

I created a list called cols containing the columns that I wanted to process.

I did not want the model to automatically treat these selected zeros as normal measurements. Instead, I converted them into missing values so that I could handle them systematically using median imputation.

2. Train-Test Split

I separated my dataset into two parts.

X contains my input features:

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age

y contains my target: Outcome

I then used `train_test_split()` to divide the dataset into training and testing portions.

Training data → 80% Testing data → 20%

3. Median Imputation

I selected the median strategy because the median is less affected by extreme values than the mean.

The imputer first calculated the median of each selected column and then replaced the missing values with the corresponding median.

This helped me preserve the patient records instead of deleting rows containing missing values.

After imputation, I checked the value counts again.

For example, SkinThickness had originally contained 227 zeros, but after preprocessing the value 29.0 appeared 244 times, showing the replacement of the missing values with the median.

Similarly, Insulin originally contained 374 zeros, and after imputation, 125.0 appeared 378 times.

4.Feature Scaling

After preparing my data, I needed to make the feature values comparable.

My features had very different numerical ranges.

For example:

- Age has relatively small values.
- Glucose has values reaching around 199.
- Insulin can have much larger values.
- DiabetesPedigreeFunction contains decimal values.

Therefore, I used StandardScaler.

for my test data.

I did not fit a separate scaler on the test data because I wanted the test data to be transformed using the same scaling parameters learned from the training data.

5. Train Logistic Regression

I selected Logistic Regression as my machine-learning algorithm.

I selected it because my target variable has two possible classes.

My model's task was essentially:

Patient information



Logistic Regression



Outcome 0 or Outcome 1

During training, the Logistic Regression model learned relationships between my input variables and the diabetes outcome. It calculated coefficients for each feature.

These coefficients later allowed me to understand which variables pushed the prediction toward or away from the positive class.

7. Evaluate the Model

I imported four evaluation metrics:

- `accuracy_score`
- `classification_report`
- `confusion_matrix`
- `roc_auc_score`

Accuracy = 75.32%

ROC-AUC = 0.8193

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.81	0.81	0.81	99
---	------	------	------	----

1	0.65	0.65	0.65	55
---	------	------	------	----

accuracy			0.75	154
----------	--	--	------	-----

macro avg	0.73	0.73	0.73	154
-----------	------	------	------	-----

weighted avg	0.75	0.75	0.75	154
--------------	------	------	------	-----

[[80 19]

[19 36]]

11. Interpret the Logistic Regression Coefficients

After evaluating the model, I wanted to understand what my Logistic Regression model had learned.

I created a new DataFrame called coefficients.

	features	coefficients
0	Pregnancies	0.206109
1	Glucose	1.030760
2	BloodPressure	-0.250151
3	SkinThickness	0.087182
4	Insulin	-0.092392
5	BMI	0.712030
6	DiabetesPedigree Function	0.218991
7	Age	0.441627

Conclusion

Overall, I successfully developed a diabetes prediction model using Logistic Regression. I cleaned and prepared the data, handled invalid values using median imputation, scaled the features, trained the model, and evaluated its performance. My model achieved 75.32% accuracy with an ROC-AUC of 0.8193. This project helped me gain practical experience in the complete machine-learning process, from data preprocessing to prediction and model interpretation.